

Impact of f(R) Gravity on Gravitational Wave-Induced Action Potential Timing Differentials in Ephaptic Axon Tracts

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Abstract

In this mathematical note, the authors consider gravitational waves under f(R) gravity. They find that due to a frequency shift towards the higher side in the f(R) case, the axon tract may have very little electrical response to these waves. f(R) gravity may effectively annul any quantum computation and quantum information processing in the nervous system. If brain quantum computing is ruled out experimentally, it might offer support to f(R)-like gravity theories which might be needed to explain dark energy.

In our 2021 publication (Journal of Applied Mathematics and Physics, Scientific Research Publishing), we demonstrated that low-frequency gravitational waves in General Relativity (GR) can measurably affect action potential timing in a tract of three ephaptically-coupled axons. The mechanism involved tracking the oscillatory displacement between spacetime-separated points—one on the axon and one along the tract axis. This displacement altered the axon's projected length onto the tract axis, leading to changes in action potential timing.

In this note, we propose a novel extension of that result to modified gravity theories, specifically f(R) gravity. Unlike GR, f(R) theories introduce an additional scalar degree of freedom due to curvature dynamics, resulting in scalar (breathing) modes in gravitational waves. These scalar modes obey a massive Klein-Gordon type equation and contribute high-frequency components to the total wave signal.

The effective waveform becomes a superposition:

$$h(t) = h_{GR} * \cos(\omega t) + \epsilon * \cos(\omega_s t) \quad (1)$$

This results in an effective upshift in frequency content. Since biological systems, including axon tracts, exhibit low-pass filtering behavior with limited action potential response bandwidths, this frequency upshift could shift the gravitational signal outside the tract's physiological response range. Consequently, under f(R) gravity, gravitational waves—despite their amplitude—may become biologically inert due to their altered spectral structure.

Thus, $f(R)$ gravity could naturally suppress gravitational wave effects on brain processing, not by weakening amplitude but by shifting frequencies beyond the neural transduction bandwidth. This provides a compelling biophysical interpretation for the absence of gravitational wave impact on neural function under extended theories of gravity.